EB-JEPA HACKATHON FIELD GUIDE

A 24-HOUR SPRINT THROUGH JOINT-EMBEDDING PREDICTIVE ARCHITECTURES

Organizing Team
EB-JEPA Hackathon
`github.com/facebookresearch/eb_jepa`

ABSTRACT

Welcome to the EB-JEPA Hackathon. Over the next 24 hours you will get hands-on with Joint-Embedding Predictive Architectures (JEPAs): self-supervised models that learn to predict in representation space rather than pixel space. This guide is your field manual. It walks you through the three working examples shipped with the `eb_jepa` library (image representation learning, video prediction, and action-conditioned world modeling with planning), shows you the small set of components you will actually edit, and hands you a curated menu of project tracks: most aimed at carrying the JEPA recipe to new continuous, high-dimensional, noisy modalities, and some at pushing the vision and robotics settings already in the codebase. It closes with the practical logistics: compute policy on the shared B200 cluster, deliverables, and how projects are judged. Read Section 1 first, skim the rest, and start running code within the hour.

1 WELCOME AND THE 24-HOUR GAME PLAN

This hackathon brings together roughly 100 participants in 25 teams of four, sharing 72 NVIDIA B200 GPUs for 24 hours. The goal is not a leaderboard win; it is to understand JEPAs by building with them. By the end you should be able to explain, and have modified, the three core pieces of every JEPA: an *encoder* f_θ that maps observations to representations, a *predictor* g_ϕ that predicts future or alternate representations, and a *regularizer* $\mathcal{R}$ that stops those representations from collapsing to a constant.

Why JEPAs, and why this codebase. Production JEPA stacks (I-JEPA, V-JEPA, V-JEPA 2) (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025) are powerful but large and hard to navigate, and world-model implementations often assume frozen pre-trained encoders and bespoke setups (Zhou et al., 2024a; Terver et al., 2026b). The `eb_jepa` library (Terver et al., 2026a) exists to close that gap: three self-contained examples, each trainable on a single GPU in a few hours, that take you from static-image self-supervision to temporal prediction to action-conditioned planning, all under one energy-based objective (Section 2). That makes it an ideal sandbox for a 24-hour sprint: you can train a model end to end several times, ablate a component, and actually see the effect before the clock runs out. By design `eb_jepa` is the prototyping rung of a ladder: once an idea works here, larger codebases pick it up, `jepa-wms` (Terver et al., 2026b) for planning with frozen encoders on diverse benchmarks, and `stable-worldmodel` (Maes et al., 2026) for a broad, controllable suite of world-model environments. Prototype small here; scale and validate there.

What we want you to explore. The library ships with vision and robotics modalities. The headline challenge of this hackathon is to take the *JEPA recipe to data it was not built for*: continuous, high-dimensional, noisy signals such as audio, physiological recordings, climate and sensor fields, financial or scientific time series, and point clouds. JEPAs are a natural fit here precisely because they predict in a learned latent space and discard task-irrelevant noise, rather than reconstructing every sample. Teams who prefer to stay in vision or robotics are very welcome to push the existing examples with innovative ideas instead; Section 6 offers a menu of both. Either way, the engineering surface is the same small set of swappable components, and Section 5 tells you exactly which files to touch.

Scope realistically. Twenty-four hours is short. A successful project is one clear hypothesis, tested with a clean before/after comparison, with a figure or number to show for it. It is far better to get one new modality training without collapsing and to *understand why* than to half-wire three ambitious ideas. The single most common failure mode in JEPA training is representation collapse; budget time for it, watch the variance and prediction losses (Section 2), and read the gotchas in each example walkthrough.

1.1 A SUGGESTED TIMELINE

Table 1 is a sane default for a team of four, not a mandate. Adapt it to your sleep schedule and your project's risk. The recurring theme: get real code running in the first hour, lock your scope early, and stop *building* with several hours to spare so you can evaluate, visualize, and write up.

Table 1: **A default 24-hour plan for a team of four.** Times are elapsed hours from kickoff. "Pods" refers to splitting the team so that environment setup, code reading, and a baseline run happen in parallel rather than in series.

Phase	Hours	What to do
Ignite	0–1	Clone, install (Section 3), launch the `image_jepa` smoke test on one GPU. Skim this guide.
Orient	1–3	In parallel pods: (i) run all three examples; (ii) read the example you will build on; (iii) pick a track (Section 6).
Lock scope	3–4	Write a one-sentence hypothesis and a success metric. Sketch the data $\rightarrow$ encoder $\rightarrow$ predictor $\rightarrow$ loss path on paper.
Baseline	4–8	Get *something* training without collapsing: data loader yields the right tensor shape, loss decreases, variance stays alive.
Iterate	8–16	One change at a time. Keep the previous run as the baseline; ablate; sweep two or three hyperparameters (Section 3).
Evaluate	16–21	Lock the model. Run downstream eval / planning / probing. Make the figures. Re-run a seed for error bars if time allows.
Write up	21–24	Slides, a short report, and a 3-minute demo. State the hypothesis, the result, and one thing you learned about JEPAs.

⋆ **Tip** Treat the first training run as a plumbing test, not a science experiment. Use a tiny model, a handful of epochs, and `logging.log_wandb=false` to confirm the pipeline runs end to end in minutes. Only then turn the knobs up. A pipeline that runs at hour 5 beats a beautiful idea that first executes at hour 20.

1.2 HOW TO READ THIS GUIDE

Section 2 is the two-page conceptual core: the unified JEPA objective and the regularizers, enough theory to act. Section 3 gets you installed and explains the launcher and the compute policy. Section 4 walks through the three examples in the run-it / read-it / tweak-it pattern. Section 5 is the reference you will return to most: the exact extension points for encoders, predictors, regularizers, and datasets. Section 6 is the project menu. Section 7 covers compute, deliverables, and judging. Throughout, `monospaced paths` point at real files in the repository, and boxes flag **tips**, **gotchas**, and **notes** worth pausing on.

2 THE JEPA RECIPE IN TWO PAGES

This section is the whole conceptual toolkit you need to start. If you remember one thing: a JEPA learns by *predicting one representation from another*, and the only reason it does not cheat by mapping everything to a constant is the regularizer. Everything else is architecture.

2.1 THE UNIFIED ENERGY OBJECTIVE

We view JEPAs through the lens of energy-based models (LeCun et al., 2006). An energy function E assigns a scalar to an input–output pair; low energy means "compatible." For a JEPA, energy is

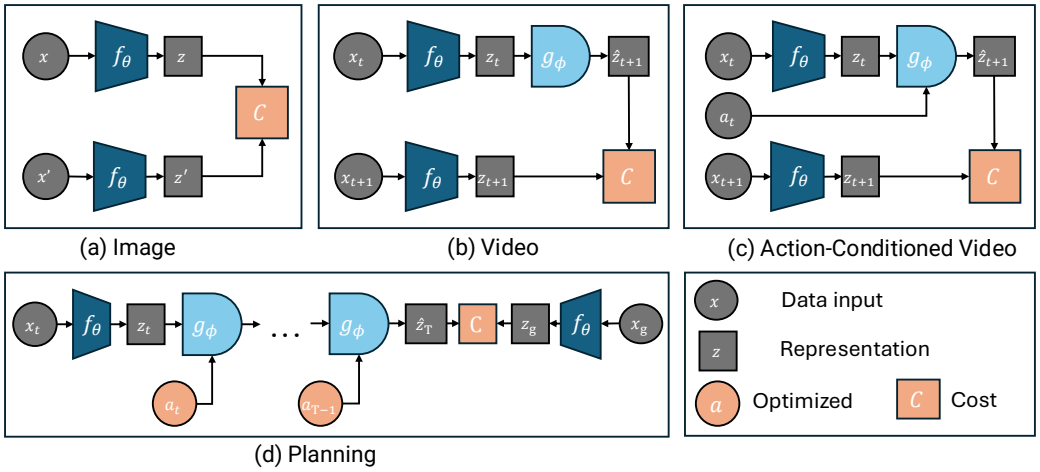

Figure 1: **One architecture, three settings.** An encoder f_θ maps inputs to representations; a predictor g_ϕ maps a context representation to a predicted target $\hat{z}$; a cost C measures prediction error in representation space. **(a)** Image: two augmented views, g_ϕ is the identity. **(b)** Video: g_ϕ predicts the next frame's representation. **(c)** Action-conditioned video: the predictor is additionally conditioned on an action a_t. **(d)** Planning: optimize an action sequence so the imagined rollout reaches a goal representation z_g.

simply prediction error in representation space. Given an encoder f_θ, a predictor g_ϕ, and optional conditioning a with its own encoder q_ω, the general energy is

$$E(x, x', a) = \mathcal{L}_{\text{pred}}\big(g_\phi(f_\theta(x), q_\omega(a)),\ f_\theta(x')\big), \tag{1}$$

and training minimizes this energy plus a regularizer $\mathcal{R}$ that prevents collapse:

$$\mathcal{L} = \mathcal{L}_{\text{pred}}\big(g_\phi(z, u), z'\big) + \lambda\,\mathcal{R}(z), \qquad z = f_\theta(x),\ u = q_\omega(a). \tag{2}$$

The three examples are Equation (2) with different choices of g_ϕ and conditioning.

(a) Image-JEPA: invariance. Two augmented views x, x' of the same image; the predictor is the identity, so the model just pulls the two representations together,

$$\mathcal{L}_{\text{image}} = \|z - z'\|_2^2 + \lambda\,\mathcal{R}(z, z'). \tag{3}$$

Low energy means the encoder has learned features invariant to the augmentation.

(b) Video-JEPA: temporal prediction. The predictor sees a context of $v{+}1$ frame representations and predicts the next one, summed over time,

$$\mathcal{L}_{\text{video}} = \sum_t \|g_\phi(z_{t-v:t}) - z_{t+1}\|_2^2 + \lambda\,\mathcal{R}(z_{1:T}). \tag{4}$$

(c) AC-video-JEPA: world modeling. An action encoder produces $u_t = q_\omega(a_{t-w:t})$ and the predictor is conditioned on it,

$$\mathcal{L}_{\text{world}} = \sum_t \|g_\phi(z_{t-v:t}, u_{t-v:t}) - z_{t+1}\|_2^2 + \lambda\,\mathcal{R}(z_{1:T}, u_{1:T}). \tag{5}$$

This is a latent dynamics model: given a state and an action, predict the next state. It is what makes planning (Section 2.4) possible.

Note. In code this is one method. `eb_jepa/jepa.py` defines a single `JEPA` module whose `unroll(...)` call handles all three settings, plus planning. Image-JEPA uses an identity predictor; video drops the action encoder; AC-video uses everything. When you build a new

modality you are picking an encoder, a predictor, and a regularizer, then handing them to this same class.

2.2 Preventing collapse: the regularizers

If you only minimized prediction error, the encoder would map everything to a constant: zero error, zero information. JEPAs in this library prevent that with explicit regularization (rather than stop-gradient or EMA tricks (Grill et al., 2020; Chen & He, 2021)). Two families are implemented.

VICReg (Bardes et al., 2022) uses two terms on a batch of embeddings $Z \in \mathbb{R}^{N\times D}$. A *variance* term keeps every feature dimension from shrinking,

$$\mathcal{L}_{\text{var}}(Z) = \frac{1}{D}\sum_{j=1}^{D} \max\Big(0,\ \gamma - \sqrt{\text{Var}(Z_{:,j}) + \epsilon}\Big), \tag{6}$$

with target standard deviation γ (typically 1); and a *covariance* term decorrelates dimensions so the model uses all its capacity,

$$\mathcal{L}_{\text{cov}}(Z) = \frac{1}{D(D-1)}\sum_{i\neq j}[C(Z)]_{i,j}^2, \qquad C(Z) = \frac{1}{N-1}(Z-\bar{Z})^\top(Z-\bar{Z}). \tag{7}$$

The full regularizer is $\mathcal{R}_{\text{VICReg}} = \alpha\mathcal{L}_{\text{var}} + \beta\mathcal{L}_{\text{cov}}$, computed on a learned projection $r = h_\psi(z)$ rather than directly on z.

SIGReg (Balestriero & LeCun, 2025) takes a different route. It identifies the isotropic Gaussian $\mathcal{N}(0, I)$ as the optimal embedding distribution and enforces it by testing Gaussianity along random 1D projections ξ_p,

$$\mathcal{R}_{\text{SIGReg}}(Z) = \frac{1}{P}\sum_{p=1}^{P} G(Z\xi_p), \tag{8}$$

where G is the Epps–Pulley Gaussianity statistic. It has a single hyperparameter λ and linear time and memory, which tends to make it more forgiving to tune than VICReg's two coefficients (Figure 2).

Gotcha. Watch the variance loss. If $\mathcal{L}_{\text{var}}$ shoots up and stays high while the prediction loss drops to near zero, your encoder is collapsing: the predictor is winning by making everything constant. The fix is almost always more regularization weight or a healthier projector, not less. The image example prints these per-term losses every epoch; the world-model example logs them too.

2.3 Two ingredients specific to world models

Multistep rollout. Training a predictor only one step ahead, then unrolling it autoregressively at test time, creates *exposure bias*: the model never saw its own predictions as input. The fix is to roll out K steps during training and sum the per-order losses,

$$\mathcal{L}_{\text{pred}} = \sum_{k=1}^{K}\mathcal{L}_k, \qquad z_{t+1}^{(k)} = g_\phi\big(z_{t-v:t}^{(k-1)},\ u_{t-v:t}\big), \quad z_t^{(0)} = f_\theta(x_{t-w:t}), \tag{9}$$

where order k counts predictor calls from a ground-truth representation. On Moving MNIST, downstream Average Precision improves steeply with K and plateaus around $K{=}4$ (Figure 3). In code, K is the `model.steps` (video) or `model.nsteps` (AC-video) config knob, consumed inside `JEPA.unroll`.

Extra world-model regularizers. Action-conditioned training in *randomized* environments needs two more terms (Sobal et al., 2022). A temporal-similarity loss encourages smooth trajectories, and an inverse-dynamics (IDM) loss (LeCun, 2022; Pathak et al., 2017) forces the representation to retain action-relevant information by predicting the action from consecutive states,

$$\mathcal{L}_{\text{sim}} = \sum_t \|z_t - z_{t+1}\|_2^2, \qquad \mathcal{L}_{\text{IDM}} = \sum_t \|a_t - \text{MLP}(z_t, z_{t+1})\|_2^2. \tag{10}$$

The full AC objective combines everything,

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \alpha\mathcal{L}_{\text{var}} + \beta\mathcal{L}_{\text{cov}} + \delta\mathcal{L}_{\text{sim}} + \omega\mathcal{L}_{\text{IDM}}. \tag{11}$$

The ablation in the paper is blunt about how load-bearing each term is: removing the IDM loss collapses Two Rooms planning from 97% to 1% success, because the encoder latches onto spurious background correlations instead of the agent.

2.4 Planning as energy minimization

Once you have a latent dynamics model, planning is optimization. To reach a goal observation x_g, search for an action sequence whose imagined rollout accumulates low energy against the goal representation,

$$E_{\text{plan}}(a_{0:H}; x_0, x_g) = \sum_{t=1}^{H} \|f_\theta(x_g) - \hat{z}_t\|_2^2, \qquad \hat{z}_{t+1} = g_\phi(\hat{z}_{t-v:t}, u_{t-v:t}),\ \hat{z}_0 = f_\theta(x_0). \tag{12}$$

Summing over *all* intermediate steps (rather than only the final state) rewards efficient paths and is robust to compounding prediction error; it beats final-state-only cost by 8 points on Two Rooms. The library solves Equation (12) with two population-based optimizers, MPPI (Williams et al., 2015) (default, soft exponential weighting over elite trajectories) and CEM (hard elite refitting), both in `eb_jepa/planning.py`. You rarely write planning code; you choose a planner config and a cost, and tune the horizon H, the number of samples N, and the number of elites Q.

3 Environment and Compute Setup

Get this working in the first hour. The detailed compute policy for the shared B200 cluster is in Section 7; here we cover installation, the launch workflow, and a smoke test.

3.1 Install

The library uses `uv` for package management. The fastest path:

```
git clone https://github.com/facebookresearch/eb_jepa.git
cd eb_jepa
uv sync                          # create .venv and install deps
source .venv/bin/activate
```

If you prefer conda for system packages:

```
conda create -n eb_jepa python=3.12 -y && conda activate eb_jepa
uv pip install -e . --group dev    # editable install +
    ↪ pytest/black/isort
```

Then point the library at your data and checkpoint directories (add these to `~/.bashrc`):

```
export EBJEPA_DSETS=/path/to/datasets      # where datasets live /
    ↪ download to
export EBJEPA_CKPTS=/path/to/checkpoints   # where runs are written
```

⋆ **Tip** Confirm the install with the test suite before you trust anything: `uv run pytest tests/`. It exercises the loss equivalences, the planning loop, and the JEPA output formats; if it passes, your environment is sane.

3.2 Run something in two minutes

Every example is a `python -m` module that reads a YAML config and accepts dotted overrides on the command line. Start with the image example and a tiny, no-logging configuration to prove the pipeline runs:

```
python -m examples.image_jepa.main \
    --fname examples/image_jepa/cfgs/default.yaml \
    optim.epochs=2 logging.log_wandb=false
```

The three entry points are:

```
python -m examples.image_jepa.main     --fname
    ↪ examples/image_jepa/cfgs/default.yaml
python -m examples.video_jepa.main     --fname
    ↪ examples/video_jepa/cfgs/default.yaml
python -m examples.ac_video_jepa.main  --fname
    ↪ examples/ac_video_jepa/cfgs/train.yaml
```

Gotcha. All configs default to `logging.log_wandb: true`. For local smoke tests set `logging.log_wandb=false` or you will hit a `wandb` login prompt. Also note datasets download to `EBJEPA_DSETS`; CIFAR-10 and Moving MNIST fetch automatically on first run (Moving MNIST is ∼800 MB and needs internet), while Two Rooms is generated on the fly.

3.3 The SLURM launcher

For anything beyond a smoke test, submit to the cluster with the shared launcher `examples/launch_sbatch.py`. It wraps single jobs, three-seed sweeps, and full hyperparameter grids.

```
# One job (dev mode)
python -m examples.launch_sbatch --example image_jepa \
    --fname examples/image_jepa/cfgs/default.yaml --single

# Three seeds {1, 1000, 10000}, wandb-averaged (recommended default)
python -m examples.launch_sbatch --example image_jepa \
    --fname examples/image_jepa/cfgs/default.yaml --sweep my_experiment

# Full grid from the YAML's sweep.param_grid section
python -m examples.launch_sbatch --example image_jepa \
    --fname examples/image_jepa/cfgs/default.yaml --sweep my_grid \
    --full-sweep --use-wandb-sweep --array-parallelism 8
```

Swap `image_jepa` for `video_jepa` or `ac_video_jepa`. Key flags: `-single` (one dev job), `-sweep NAME` (name your run group), `-full-sweep` (read `sweep.param_grid` from the config), `-use-wandb-sweep` (the wandb sweep UI), and `-array-parallelism N` (cap concurrent jobs, which matters a lot on a shared cluster, see Section 7).

Gotcha. The launcher's default SLURM settings (partition, account, memory) at the top of `examples/launch_sbatch.py` point at the original authors' cluster. The organizers will give you the correct `-partition`/`-account` (or a patched launcher) for the hackathon B200 nodes. Do not burn an hour debugging a queue rejection; ask. See Section 7.

3.4 Where runs land, and how to compare them

Runs are written under `$EBJEPA_CKPTS/{example}/` with an auto-named experiment folder encoding the key hyperparameters, e.g. `resnet_vicreg_proj_bs256_ep300_std1.0_cov80.0_seed1`. Three-seed sweeps share a wandb run *name* so you can group by name in the UI to get mean and standard error automatically. To compare a before/after change, change *one* thing, keep the same seed set, and read the matched-epoch metric.

⋆ **Tip** The recommended scientific unit is the three-seed sweep, not a single run. JEPA training has real seed variance, and a one-point "improvement" from a single run is usually noise. With B200s you can afford three seeds; budget for it.

4 The Three Examples: Run It, Read It, Tweak It

The library is built around one idea: the training loop, the rollout, and the losses are modality-agnostic and live in the shared `eb_jepa/` package; each example in `examples/` is a thin `main.py` that assembles an encoder, a predictor, and a regularizer, then calls `JEPA.unroll`. Once you have read one example, you have read them all. We go through them in order of increasing complexity. For each: how to run it, the handful of files worth reading, the config knobs that matter, and what to tweak first.

Note. There is no central "registry" of encoders or losses. Components are plain `nn.Module`s wired together by hand inside each example's `main.py`, sometimes behind a small `if cfg.model.type == "..."` branch. To add your own, you edit the builder in `main.py` (and occasionally add a class to `eb_jepa/architectures.py` or `eb_jepa/losses.py`). Section 5 is the map.

4.1 Image-JEPA: self-supervised features on CIFAR-10

What it teaches. The purest form of the recipe: no predictor, no time. Two augmented views of an image, pull their representations together, and keep them from collapsing with VICReg or SIGReg. A linear probe trained online reports classification accuracy each epoch (∼91% with SIGReg, ∼90% with VICReg at 300 epochs).

Run it.

```
# ResNet-18 + VICReg (default)
python -m examples.image_jepa.main --fname
    ↪ examples/image_jepa/cfgs/default.yaml
# ResNet-18 + SIGReg
python -m examples.image_jepa.main --fname
    ↪ examples/image_jepa/cfgs/sigreg.yaml
# ViT-S + VICReg
python -m examples.image_jepa.main --fname
    ↪ examples/image_jepa/cfgs/transformers.yaml
```

There is no separate eval step: the probe accuracy `val_acc` is logged every epoch during pretraining.

Read it.

- `examples/image_jepa/dataset.py` returns two augmented views per image (crop, color jitter, grayscale, solarize, flip).
- `examples/image_jepa/main.py`: encoder built at lines `435-468`; loss chosen at `518-522`; the batch loop with the online probe at `275-321`; the epoch loop at `539-605`.
- `eb_jepa/losses.py`: `VICRegLoss` and `BCS` (the SIGReg implementation) take two views and return a loss dict.

Key knobs (`cfgs/default.yaml`).

```
model:  {type: resnet, use_projector: true, proj_hidden_dim: 2048,
    ↪ proj_output_dim: 2048}
loss:   {type: vicreg, std_coeff: 1.0, cov_coeff: 80.0, lmbd: 10.0}  #
    ↪ lmbd used iff type=bcs
optim:  {epochs: 300, lr: 0.3, weight_decay: 1.0e-4}                  #
    ↪ LARS + cosine warmup
data:   {batch_size: 256}
```

Tweak it first.

- Swap the regularizer: `loss.type=bcs loss.lmbd=10`. SIGReg has one coefficient and is more forgiving (Figure 2).
- Swap the backbone: `model.type=vit_s`. Compare convergence and final accuracy at equal epochs.
- Ablate the projector: `model.use_projector=false`. Expect a $\sim$3 point accuracy drop, a clean demonstration of why the regularizer is computed in a projected space.

Gotcha. Bad regularization weights collapse this model fast. The paper's sweep shows `std=100, cov=100` dropping VICReg to 10% (chance) while `std=1, cov=100` reaches 90%. If `val_acc` is stuck near 10%, you are collapsed, not under-trained. Also: the ViT `patch_size` is currently hardcoded in `main.py`, so the `model.patch_size` YAML key is ignored for ViT.

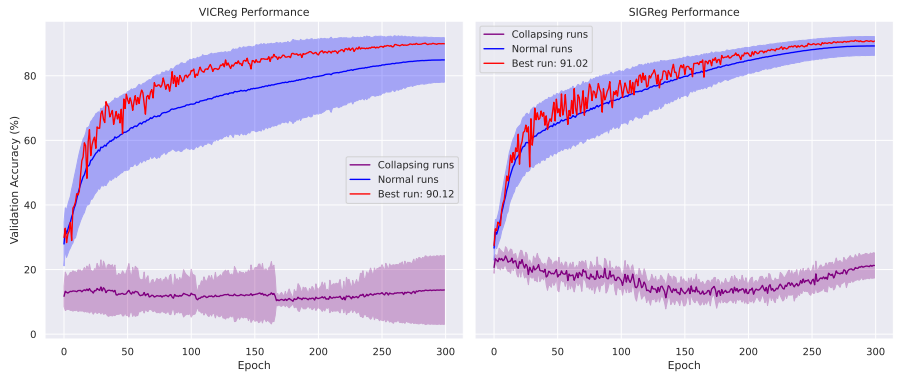

Figure 2: **Hyperparameter sensitivity on CIFAR-10.** SIGReg (right) stays stable across a naive sweep while VICReg (left) reaches a similar peak but needs more careful tuning. Collapsed runs flatline near chance accuracy. Use this as your mental model when a new run will not improve: suspect collapse before suspecting capacity.

4.2 Video-JEPA: Temporal Prediction on Moving MNIST

What it teaches. The predictor wakes up. Frames are encoded, the past representations are context, and the model predicts the next frame's representation, unrolled K steps to fight exposure bias. A detection head and a pixel decoder (both trained on detached features, purely for evaluation) let you score Average Precision and watch the rollout.

Run it.

```
python -m examples.video_jepa.main --fname
    ↪ examples/video_jepa/cfgs/default.yaml
# fewer rollout steps, larger batch:
python -m examples.video_jepa.main --fname
    ↪ examples/video_jepa/cfgs/default.yaml \
    model.steps=2 data.batch_size=128
```

Read it.

- `eb_jepa/datasets/moving_mnist.py`: `MovingMNISTDet`, two bouncing digits; the dataset, not the example, owns the data.
- `examples/video_jepa/main.py`: model assembled at `128-142` (`ResNet5` encoder, `ResUNet` predictor wrapped by `StateOnlyPredictor`, `VCLoss`, `SquareLossSeq`); training step at `191-204`.

- `eb_jepa/jepa.py`: `unroll` in `parallel` mode, the K-step loop, at `142-157`. This is the heart of temporal training.

Key knobs (`cfgs/default.yaml`).

```
model: {dobs: 1, henc: 32, dstc: 16, hpre: 32, steps: 4}   # steps = K
    ↪ rollout depth
loss:  {std_coeff: 10.0, cov_coeff: 100.0}
optim: {epochs: 50, lr: 1.0e-3}
data:  {batch_size: 64}
```

Tweak it first.

- Sweep the rollout depth `model.steps` $\in \{1, 2, 4, 8\}$ and plot AP versus prediction horizon (Figure 3). This reproduces the paper's central video result in an afternoon.
- Replace the spatial `ResUNet` predictor with the GRU `RNNPredictor` (already in `eb_jepa/architectures.py`) and compare.

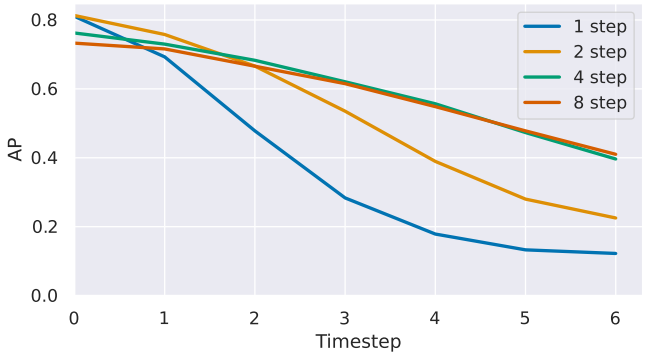

Figure 3: **Multistep rollout helps, and saturates.** Average Precision of the autoregressive prediction versus horizon, for models trained with $K \in \{1, 2, 4, 8\}$ rollout steps. Training with longer rollouts aligns training with autoregressive inference; the Pareto knee is around $K{=}4$. A clean, cheap result to reproduce or extend to a new temporal modality.

4.3 AC-Video-JEPA: A world model you can plan with

What it teaches. The full stack: an action-conditioned latent dynamics model, the two world-model regularizers ($\mathcal{L}_{\text{sim}}$ and the critical $\mathcal{L}_{\text{IDM}}$), and goal-conditioned planning by energy minimization. The agent navigates a "Two Rooms" environment whose wall and door are randomized per trajectory, so the task is non-monotonic (sometimes you must move *away* from the goal to reach it). The best model plans at 97% success with MPPI.

Run it (train, then plan).

```
# Train the world model
python -m examples.ac_video_jepa.main --fname
    ↪ examples/ac_video_jepa/cfgs/train.yaml
# Evaluate planning on a trained checkpoint (runs unroll + planning
    ↪ eval)
python -m examples.ac_video_jepa.main \
    --meta.model_folder /path/to/run --meta.eval_only_mode True
```

Planning uses MPPI by default (`cfgs/planning_mppi.yaml`); point `eval.plan_cfg_path` at `cfgs/planning_cem.yaml` for CEM.

Read it.

- `eb_jepa/datasets/two_rooms/`: the environment (`DotWall`), the dataset, and the data-generation pipelines (online / streaming / offline).

- `examples/ac_video_jepa/main.py`: model built at `176-230` (`ImpalaEncoder`, `RNNPredictor`, action encoder `nn.Identity()`, `InverseDynamicsModel`, `VC_IDM_Sim_Regularizer`); training step calling `jepa.unroll(..., unroll_mode="autoregressive")` at `334-343`.
- `eb_jepa/planning.py`: `CEMPlanner` and `MPPIPlanner`, the goal-distance objective `ReprTargetDistMPCObjective`, and the eval entry points `main_eval` / `main_unroll_eval`.

Key knobs.

```
# cfgs/train.yaml
model:  {encoder_architecture: impala, dobs: 2, dstc: 32, nsteps: 8}  #
    ↪ nsteps = K
  regularizer: {cov_coeff: 8, std_coeff: 16, sim_coeff_t: 12,
    ↪ idm_coeff: 1}
optim:  {epochs: 12, lr: 0.001}
data:   {batch_size: 384}
# cfgs/planning_mppi.yaml
planner: {planner_name: mppi, plan_length: 90, n_iters: 20,
    ↪ num_samples: 200, num_elites: 20, temperature: 0.005}
```

Tweak it first.

- Reproduce the headline ablation: set `model.regularizer.idm_coeff=0` and watch planning success collapse toward 1%. Then zero `std_coeff`, `cov_coeff`, `sim_coeff_t` one at a time.
- Compare planners (MPPI vs CEM) and the cost design (`planning_objective.sum_all_diffs` true vs false, i.e. cumulative vs final-state cost). Cumulative is worth ∼8 points.
- Trade planning compute for success: sweep `num_samples` and `plan_length` and report success-versus-wall-clock.

Gotcha. The action dimension `action_dim=2` is hardcoded in several places (`main.py`, `InverseDynamicsModel`, the planners). If you retarget this example to a new control problem, grep for `action_dim` and change all of them together. And remember: in randomized environments the IDM loss is what keeps the encoder honest, do not drop it.

5 Extension Points: The Files You Will Actually Edit

This is the reference you will come back to. Extending the library is not mysterious: there is no plugin system to learn. You add a plain `nn.Module` and wire it into an example's `main.py` builder. Table 2 is the "I want to… → edit this" map; the rest of the section explains the two contracts that matter (the encoder/predictor I/O shape, and the regularizer return signature).

5.1 The Central Object: `JEPA.unroll`

Everything funnels through one class, `JEPA` in `eb_jepa/jepa.py`, constructed as `JEPA(encoder, action_encoder, predictor, regularizer, predcost)`. Its `unroll(observations, actions, nsteps, unroll_mode, ...)` method does, in order: encode the observations to `state`; if computing loss, call `regularizer(state, actions)`; encode the actions; then roll the predictor forward and accumulate the prediction cost. Two rollout modes:

- `unroll_mode="parallel"` (used by `video_jepa`): convolutional predictors predict all timesteps at once, refeeding a few ground-truth context frames; the K-step loop is at `jepa.py:142-157`.
- `unroll_mode="autoregressive"` (used by `ac_video_jepa` and any RNN predictor): step-by-step with a sliding context window; loop at `jepa.py:161-190`.

Table 2: **Where to make each kind of change.** "Builder" means the model assembly block near the top of the relevant `examples/*/main.py`.

I want to…	Edit
Add an encoder for a new modality	New `nn.Module` in `eb_jepa/architectures.py`; wire it into the builder (`image_jepa/main.py`∼435, or the hardcoded constructor in `video_jepa/main.py`∼130 / `ac_video_jepa/main.py`∼186).
Add a predictor	New `nn.Module` in `architectures.py` exposing `is_rnn` and `context_length`; wire into the builder.
Add / change a regularizer	New class in `eb_jepa/losses.py`; instantiate in the builder, or add an `elif cfg.loss.type` branch (image).
Add a dataset / modality	New `Dataset` (or `TrajDataset`) under `eb_jepa/datasets/`; for the world-model path, add an `env_name` branch in `eb_jepa/datasets/utils.py:init_data` (it currently rejects anything but `two_rooms`); for the image path, edit the dataset block in `image_jepa/main.py`∼398.
Add a planner or planning cost	New class in `eb_jepa/planning.py` plus an entry in `planner_name_map` / `objective_name_map`.
Change rollout depth, coefficients	Config only: `model.steps`/`model.nsteps`, `std_coeff`, `cov_coeff`, `sim_coeff_t`, `idm_coeff`, `loss.lmbd`.

`JEPA` chooses behavior by reading `getattr(predictor, "is_rnn", False)` and `predictor.context_length`. So your predictor mostly needs to declare those two attributes correctly and the rest follows.

5.2 Contract 1: encoder and predictor tensor shapes

The temporal examples speak in 5D tensors `[B, C, T, H, W]` (batch, channels, time, height, width). The mixin `TemporalBatchMixin` in `eb_jepa/nn_utils.py` folds time into the batch for you, so a 2D encoder can be written as if it only ever sees `[B, C, H, W]`; subclass it and implement `_forward`. Encoders output a representation `[B, D, T, H', W']` (the `ImpalaEncoder` pools to `H'=W'=1`, i.e. one vector per frame). A predictor implements `forward(state, action) -> [B, D, T, ...]` and declares:

```
class MyPredictor(nn.Module):
    is_rnn = False          # True -> autoregressive single-step unroll
    context_length = 2      # how many context frames it consumes
    def forward(self, state, action):
        ...                 # return predicted next-step
    ↪ representation(s)
```

⋆ **Tip** For a new modality with no spatial grid (a 1D time series, a feature vector per step), the cleanest path is the GRU `RNNPredictor` (it sets `is_rnn=True`, `context_length=0`) plus an encoder that outputs `[B, D, T, 1, 1]`. You inherit the autoregressive rollout and planning for free. Reshape your samples to that 5D convention and most of the spatial machinery becomes a no-op.

5.3 Contract 2: the regularizer return signature

There are two regularizer conventions, matching the two settings:

- **Sequence regularizers** (video / world model) accept `(state, actions)` and return a triple `(weighted_loss, unweighted_loss, loss_dict)`. This is what `JEPA.unroll` expects. Models: `VCLoss` (variance+covariance) and `VC_IDM_Sim_Regularizer` (the full world-model regularizer).

- **Two-view SSL losses** (image) accept `(z1, z2)` and return a dict with a `"loss"` key plus per-term entries. Models: `VICRegLoss` and `BCS` (SIGReg). The image training loop logs every extra key in the dict automatically, so new sub-losses show up in logging with no extra work.

Match whichever contract fits your setting and the rest of the loop is untouched. Building blocks you can reuse to compose a new regularizer live in `eb_jepa/losses.py`: `HingeStdLoss` (variance), `CovarianceLoss`, `TemporalSimilarityLoss`, `InverseDynamicsLoss`, and the `BCS` Gaussianity statistic.

5.4 Adding a dataset: the one real dispatch

For image-style two-view data, edit the dataset block in `image_jepa/main.py` (around line 398) to branch on `cfg.data.dataset` and adjust `num_classes` and the normalization constants. For the world-model path, `eb_jepa/datasets/utils.py:init_data` is the single point that maps an `env_name` to data loaders, and it currently raises for any `env_name` other than `two_rooms`. Adding a new environment or trajectory dataset means implementing a `TrajDataset` (which yields `(obs, action, state, ...)` slices via `TrajSlicerDataset`) and adding your branch there. This is the most involved extension, so scope it early.

Gotcha. Per-feature normalization is not optional for new modalities. The variance term in VICReg measures per-dimension standard deviation; if one input channel dwarfs the others (common in EEG, audio, sensor fusion), it will dominate both the encoder and the regularizer. Normalize each channel/feature (z-score or robust scaling) before it enters the encoder.

6 Project Tracks

Below is a menu, not a syllabus. Pick one track, or use one as a springboard for your own idea. Each track names a difficulty, the example it forks, a 24-hour goal, the files you will touch, and a success metric. Difficulty badges: **Warm-up** is reachable by most teams and a safe first project; **Standard** is a solid full-day project; **Ambitious** is high-risk, high-reward, attempt it only if your baseline is already running.

What makes a good new-modality target. The headline theme is *continuous, high-dimensional, noisy* data, exactly where predicting in pixel or sample space is wasteful and predicting in a learned latent space shines. A good target has: (i) abundant unlabeled data for pretraining; (ii) a small labeled set for a downstream probe; (iii) temporal or multi-view structure the predictor can exploit; and (iv) enough noise that reconstruction would be a bad idea. EEG (Track 1) is the worked example of all four.

6.1 Group A: Carry the JEPA recipe to a new modality

Track 1 (Flagship) — EEG abnormality detection on TUAB — Standard / Ambitious

The idea. Pretrain a JEPA on raw clinical EEG, then linear-probe it to classify recordings as normal vs abnormal. This mirrors the EEG foundation model LaBraM (Jiang et al., 2024) but replaces its VQ tokenizer and Fourier-spectrum reconstruction with a pure joint-embedding objective: *predict patch representations in latent space*, never the noisy raw signal. That structural choice is the point. LaBraM had to reconstruct the spectrum because raw-EEG reconstruction would not converge; a JEPA sidesteps the problem entirely, since the target encoder is free to discard unpredictable artifacts.

Data. TUH Abnormal EEG Corpus (TUAB), a labeled subset of the TUH EEG Corpus (Obeid & Picone, 2016): ∼2,717 training and 276 evaluation recordings, binary normal/abnormal, in EDF format. Pretrain on a small slice of the larger unlabeled TUH corpus (tens to ∼100 hours is plenty for 24h; do not try to ingest the full corpus). Standardize to a common ∼21–23 channel montage at 200 Hz, bandpass 0.1–75 Hz, 60 Hz notch, z-score per channel.

Two framings (start with the first).

- *Temporal* (video-JEPA style, Section 4.2): cut each channel into 1-second patches; encode the context window; predict the *next* 1-second window's representation across all channels. This is next-frame prediction with frames = time windows.
- *Masked* (image-JEPA style, Section 4.1): build the channel×time patch grid, mask a fraction of patches, predict their representations from the visible ones; use EEG augmentations (channel dropout, amplitude scaling, time jitter) for the two-view variant.

Files to touch. A new `Dataset` under `eb_jepa/datasets/` that loads EDF (use `MNE-Python` or `braindecode`'s `TUHAbnormal`) and yields `[B, C, T, 1, 1]` patch tensors; a small conv-patch or 1D-conv encoder in `architectures.py`; the GRU `RNNPredictor` for the temporal framing; a linear probe modeled on `image_jepa/eval.py`.

Metric. Balanced Accuracy on the 276-recording eval set (also report AUROC). Context: supervised CNNs reach ∼0.79 Balanced Accuracy, BIOT ∼0.796, fine-tuned LaBraM-Base ∼0.81. A frozen JEPA *linear probe* clearing ∼0.78 would be a genuinely strong hackathon result.

Gotchas. Variable montages across recordings (restrict to the common 10–20 set for simplicity, or use per-electrode embeddings); per-channel normalization is mandatory; class-aware metric because the eval split is mildly imbalanced. Report metrics at the recording level, not the raw 10-second-window level.

Track 2 — Audio / speech representation JEPA — Standard

Goal. Self-supervised audio features, probed on keyword spotting (Speech Commands) or environmental sound (ESC-50/UrbanSound). Treat log-mel spectrograms as single-channel "images" for the image-JEPA masked/augmented setting, or treat mel frames as a temporal sequence for the video-JEPA setting. **Files:** a spectrogram dataset; reuse the ResNet or ViT encoder. **Metric:** linear-probe accuracy on the downstream classification set. **Why JEPA:** audio is continuous and noisy; latent prediction avoids modeling phase and background noise.

Track 3 — Wearable biosignals: ECG / IMU / PPG — Warm-up / Standard

Goal. JEPA on accelerometer/gyroscope or ECG streams; probe on Human Activity Recognition (UCI-HAR, PAMAP2) or arrhythmia (PhysioNet). These are low-spatial multivariate time series, the ideal fit for the GRU predictor and `[B, C, T, 1, 1]` convention. **Files:** a windowed time-series dataset; 1D-conv encoder; `RNNPredictor`; linear probe. **Metric:** probe accuracy / F1. The lowest-friction new-modality track, good for a first-time team.

Track 4 — Latent dynamics of physical fields (The Well) — Standard / Ambitious

Goal. The Well (Ohana et al., 2024) is a 15 TB collection of 16 physics-simulation datasets (fluid dynamics, magneto-hydrodynamics, astrophysics, acoustics, active matter) stored as spatiotemporal fields, the continuous, high-dimensional modality JEPAs were made for, with a ready-made PyTorch loader. Train a video-JEPA-style model to predict the next field state in latent space, then answer the open question: does latent prediction give more stable long-horizon rollouts than the field-space neural-operator surrogates (FNO, U-Net) that The Well ships as baselines? **Files:** wrap `the_well.data.WellDataset` to emit `[B, C, T, H, W]` clips; reuse the `ResUNet` spatial predictor and `VCLoss`; add a small decoder from latent back to the field for scoring. **Data.** Start at the small end of the 6.9 GB–5.1 TB range, a 2D set such as `active_matter` or `gray_scott_reaction_diffusion`; do not pull the multi-TB sets. **Metric:** VRMSE of the decoded autoregressive rollout vs the FNO/U-Net baseline at several horizons. A clean JEPA-versus-surrogate comparison on real physics (for *observational* rather than simulated fields, ERA5/WeatherBench is a harder variant).

Track 5 — Scientific / financial multivariate time series Standard

Goal. JEPA on noisy multivariate series (sensor networks, traffic, market data) with a downstream forecasting or regime-classification probe. The interesting research question: does latent prediction with anti-collapse regularization learn more transferable features than direct forecasting on noisy series? **Files:** time-series dataset; 1D encoder; `RNNPredictor`. **Metric:** downstream probe vs a supervised baseline on the same split.

Track 6 — 3D point clouds / LiDAR Ambitious

Goal. View-invariant JEPA on point clouds (ModelNet/ShapeNet): two augmented samplings/rotations of an object are the two views (image-JEPA setting), probed on shape classification. **Files:** a point-cloud dataset with geometric augmentations; a PointNet-style encoder in `architectures.py`; reuse `VICRegLoss`. **Metric:** linear-probe classification accuracy. Tests whether the recipe transfers to an unordered, irregular modality.

6.2 Group B: Push the vision and robotics examples

Track 7 — Learned cost / value for planning Standard / Ambitious

Goal. Distance-in-latent-space is a crude planning cost. Replace it with a learned cost or value function (TD-MPC style (Hansen et al., 2024; 2022)) trained on the world model's own rollouts, so the agent optimizes a quantity that correlates with task success rather than raw representation distance. **Files:** a new objective class in `eb_jepa/planning.py` registered in `objective_name_map`; a small value head trained alongside the JEPA. **Metric:** Two Rooms success and planning compute vs the distance-cost baseline, especially on the hardest non-monotonic episodes. Attacks a limitation the paper explicitly calls out as open.

Track 8 — Hierarchical / multi-timescale world model Ambitious

Goal. JEPAs here predict at a single temporal resolution; intelligent planning wants several. Add a coarse predictor that models long-horizon abstractions on top of the fine per-step predictor, and plan with the coarse level to cut the search horizon. The paper names its modular encoder/predictor/regularizer split as "a natural starting point" for exactly this, and no example ships it yet. **Files:** a second predictor and a two-level unroll in a forked `ac_video_jepa/main.py`; a coarse-level temporal-similarity term. **Metric:** long-horizon planning success and wall-clock vs the flat baseline. The deepest open robotics direction in the guide.

Track 9 — Stress-test the recipe under factors of variation Ambitious

Goal. `eb_jepa`'s Two Rooms is deliberately tiny. The open question is whether the *minimal* recipe (VC + sim + IDM, distance-cost MPPI) survives realistic perturbations, and which term breaks first. Train an AC-video-JEPA-style model and evaluate planning as you dial up controllable visual, geometric, and physical factors of variation, using the environment suites in `stable-worldmodel` (Maes et al., 2026) or the diverse planning benchmarks in `jepa-wms` (Terver et al., 2026b) rather than hand-building a new environment. **Files:** adapt one of those suites' envs to the `eb_jepa` JEPA/planning stack (encoder + `JEPA.unroll` + a planner). **Metric:** a success-vs-perturbation curve against the Two Rooms baseline, and which regularizer term must be strengthened as the world gets harder. Ambitious but high-signal: it tells you where the toy recipe actually ends.

Track 10 — Does intuitive physics emerge? Standard

Goal. Recent work shows physical intuition emerges from self-supervised video pretraining (Garrido et al., 2025). Test it in miniature on the video-JEPA: feed it physically plausible vs impossible Moving MNIST sequences (a digit teleporting, passing through a wall, or reversing instantly) and ask whether prediction energy in latent space spikes on the impossible ones, a violation-of-expectation signal. **Files:** a probe that constructs paired plausible/impossible clips and compares `predcost` energy; light additions to the video eval. **Metric:** the energy gap (impossible > plausible) and how it grows over training. Exploratory, cheap, and genuinely open at this scale.

⋆ **Tip** Whatever you pick, the deliverable that wins is a *controlled comparison*: a baseline and one change, three seeds each, one figure. "We ported JEPA to EEG and the linear probe hits 0.79 Balanced Accuracy, and here is the collapse we hit at zero covariance weight and how we fixed it" beats a pile of half-finished ideas.

7 Compute, Logistics, and Judging

7.1 The compute budget

We share **72 NVIDIA B200 GPUs** across **25 teams** for 24 hours. That is about **2.9 GPUs per team** on average. The good news: every example in this library is designed for *single-GPU* training and finishes in a few hours, so a team's natural unit of work is "a few single-GPU jobs in parallel," not one giant distributed run. A three-seed sweep is three single-GPU jobs; a small hyperparameter grid is a handful more.

Note. The numbers and the SLURM details below are the organizers' *proposed* policy. The exact `-partition`, `-account`, per-GPU memory, and any fair-share quotas will be confirmed at kickoff and pinned in the event channel. If a launch is rejected by the scheduler, it is almost certainly a partition/account mismatch, ask in the channel rather than debugging it alone.

Fair-share rules of thumb.

- Prefer many *single-GPU* jobs over multi-GPU jobs. The examples do not need more than one GPU, and single-GPU jobs schedule faster and waste less.
- Cap your concurrency with `-array-parallelism`. A sane ceiling is **3 concurrent GPUs per team** during the day; if the queue is empty overnight, use more, but yield when teams are waiting.
- Kill dead runs. A collapsed run (flat variance, chance accuracy) will not recover; cancel it and free the GPU rather than letting it burn 12 hours.
- Smoke-test on `-single` with `optim.epochs` tiny before launching a sweep. A typo that crashes at epoch 0 across 12 array tasks wastes everyone's slots.

Disk and checkpoints. Runs write to `$EBJEPA_CKPTS`. Checkpoints add up fast across a sweep; keep only the best and the latest, and clean up smoke tests. Datasets live in `$EBJEPA_DSETS`; download a shared copy once per node rather than per team where possible (the organizers will pre-stage CIFAR-10, Moving MNIST, and any agreed new-modality datasets).

7.2 Notes on B200 vs the default H100 configs

The shipped configs are tuned for H100s. B200s have substantially more memory and strong `bfloat16` throughput, so the defaults will run but leave performance on the table. Practical adjustments:

- **Increase batch size.** The image example uses `batch_size=256` and the world model `384`; you can likely push these up on a B200. Scale the learning rate accordingly and re-check for collapse.

- **Keep `bfloat16`.** The configs already set `use_amp: true` with `bfloat16` (image, world model) or `float16` (video). Prefer `bfloat16` on B200; if you see NaNs under `float16`, switch the video config to `bfloat16`.
- **Try `torch.compile`.** The world-model example already supports it; compilation pays off most on the longer temporal rollouts.
- **Do not over-scale.** A bigger batch is not free accuracy; for SSL it changes the regularization dynamics. Treat any batch-size change as an ablation with its own before/after, not a default win.

⋆ **Tip** The bottleneck for these small models is often the data loader, not the GPU. If GPU utilization is low, raise `data.num_workers` and enable `persistent_workers`/`pin_mem` before reaching for a bigger model. For a new modality, a slow EDF/NetCDF reader can starve a B200 completely; pre-process to a fast on-disk format (memmap, `.npy`, webdataset) once, up front.

7.3 Deliverables

By the 24-hour mark each team submits:

1. **A code artifact**: a fork or branch with your changes, runnable from a single documented command, plus the config(s) you used.
2. **A short report** (1–2 pages or a slide deck): the hypothesis, the setup, the result with at least one figure or table, and one paragraph on what you learned about JEPAs (especially any collapse you fought).
3. **A 3-minute demo** to the room: ideally a live or recorded artifact, a probe-accuracy curve, a planning rollout GIF, a reconstruction-free latent prediction, whatever shows your result.

7.4 How projects are judged

Table 3 is the rubric. Note what is *not* on it: raw leaderboard position. A negative result, cleanly shown ("latent prediction did not beat the supervised baseline on this modality, and here is the evidence and our hypothesis"), scores well. Confident hand-waving does not.

Table 3: **Judging rubric.** Equal weight on understanding and execution; creativity and clarity break ties.

Criterion	What we look for
JEPA understanding	Can the team explain encoder/predictor/regularizer, why collapse happens, and how their design avoids it?
Scientific rigor	A clear hypothesis and a controlled before/after comparison; seeds and error bars where feasible.
Result	A real, interpretable outcome (positive or negative) with a figure/number, not just "it ran."
Creativity	A modality or idea that genuinely stresses the recipe, or an elegant solution to a collapse/scaling problem.
Clarity	A report and demo a non-expert teammate could follow.

7.5 Getting unstuck

Read the gotcha boxes in Section 4 first; most early problems are there. For collapse, re-read Section 2 and watch the per-term losses. For anything cluster-related, use the event channel. Mentors will run office hours on a posted schedule. And remember the meta-advice: lock scope early, get a pipeline running in hour five, and stop building in time to measure. Have a great 24 hours.

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